5.8 Stokes' Theorem_Guide

Monday, January 12, 2026 10:44 PM

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Theorem...

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Definition: Suppose that $\mathbf{F}(x,y,z)$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of the components of $\mathbf{F}$ exist. The <u>curl of F</u> is the vector field on $\mathbb{R}^3$ defined by

$$\operatorname{curl}(\mathbf{F}) = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}.$$

Remark: To help ourselves in remembering the formula for curl, we may write $\operatorname{curl}(\mathbf{F}) = \vec{\nabla} \times \mathbf{F}$, since

$$\vec{\nabla} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, -\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right), \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle$$

<u>Example 1:</u> Find the curl of the vector field: $\mathbf{F}(x,y,z) = \langle \sin(yz), \sin(zx), \sin(xy) \rangle$.

$$\operatorname{Curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin(yz) & \sin(zx) & \sin(xy) \end{vmatrix}$$

$$= \langle \cos(xy)x - x\cos(zx), -\cos(xy)y + \cos(yz)y, \cos(zx)z - \cos(yz)z \rangle$$

<u>Example 2:</u> Suppose that $f(x,y,z)$ is a function of three variables that has continuous second-order partial derivatives. Find a formula for $\operatorname{curl}(\vec{\nabla} f)$.

Clearly $\vec{\nabla} f = \langle f_x, f_y, f_z \rangle$

$$\text{So } \operatorname{curl}(\vec{\nabla} f) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_x & f_y & f_z \end{vmatrix} = \langle f_{zy} - f_{yz}, -f_{zx} + f_{xz}, f_{yx} - f_{xy} \rangle$$

$$= \langle 0, 0, 0 \rangle$$ Since All mixed partials are equal due to continuous 2nd partials

$$= \vec{0}$$

The above example leads to the following result.

Theorem: If the vector field $\mathbf{F} = \langle P, Q, R \rangle$ is conservative and P, Q, and R have continuous first-order partial derivatives, then $\text{curl}(\mathbf{F}) = \mathbf{0}$.

Example 3: Show that the vector field $\mathbf{F}(x,y,z) = (xz)\mathbf{i} + (xyz)\mathbf{j} - (y^2)\mathbf{k}$ is not conservative.

$$\text{Curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} = \langle -2y - xy, x, yz \rangle \neq \vec{0}$$

Since $\vec{F}$ has cont. 1^{st} partials & $\text{curl}(\vec{F}) \neq 0$ then

$\vec{F}$ is <u>NOT</u> conservative.

Theorem: If $\mathbf{F}$ is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and $\text{curl}(\mathbf{F}) = \mathbf{0}$, then $\mathbf{F}$ is a conservative vector field.

Example 4: Consider the vector field: $\mathbf{F}(x,y,z) = \langle 1, \sin(z), y\cos(z) \rangle$.

a) Show that $\mathbf{F}$ is a conservative vector field.

$$\text{Curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 1 & \sin(z) & y\cos(z) \end{vmatrix} = \langle \cos(z) - \cos(z), -0, 0 \rangle$$

$$= \vec{0}$$

Since the 1^{st} partials of $\vec{F}$ are cont. & $\text{curl}(\vec{F}) = \vec{0}$ then $\vec{F}$ is conservative.

b) Find a potential function for $\mathbf{F}$.

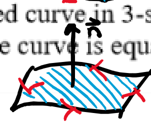
let $f(x,y,z) = x + y\sin(z)$

Application: Suppose that $\mathbf{F}$ represents the velocity field in fluid flow. Particles near (x,y,z) in the fluid tend to rotate about the axis that points in the direction of $\text{curl}(\mathbf{F}(x,y,z))$, and the length of this curl vector is a measure of how quickly the particles move around the axis. If $\text{curl}(\mathbf{F}(x,y,z)) = \mathbf{0}$ at a point P we say that the fluid is free from rotations at P (i.e. there is no whirlpool or eddy at P) and $\mathbf{F}$ is called <u>irrotational</u> at P.

Recall: We had two ideas for how to generalize Green's Theorem:

1. We might try to increase the "dimensions of the integrals" in Green's Theorem by one. More precisely, we might guess that the flux of a vector field through a closed surface is equal to a triple integral taken over the region enclosed by the surface.

2. Alternatively, if we have a closed curve in 3-space (rather than 2-space), we might guess that the line integral of a vector field over the curve is equal to a surface integral taken over the region that the curve encloses.



C σ

In this section we purse the second of these ideas. However, before stating Stokes' Theorem, we need a definition.

> **Definition:** Suppose that curve C is the boundary curve of some surface σ in $\mathbb{R}^3$. Suppose that the unit normal vector $\mathbf{n}$ determines an orientation for σ. The orientation of σ induces the positive orientation of C in the following sense. If one walks around C with his or her head in the direction of $\mathbf{n}$ and σ is always on the left, then one is following the positive orientation of C.

Note: There is a nice illustration of this definition on page 1134 of the textbook.

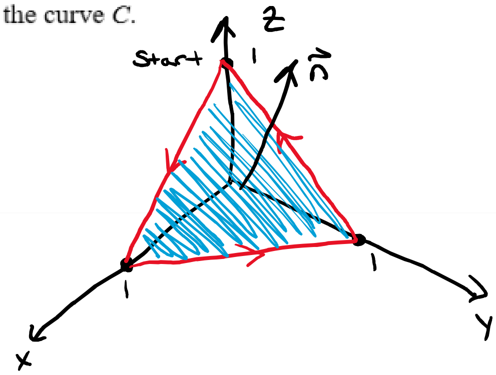
> **Theorem (Stokes' Theorem):**
> Let σ be an oriented piecewise-smooth surface that is bounded by a simple, closed piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains σ. Then, $\int_C \mathbf{F}\cdot d\mathbf{r} = \iint_\sigma \operatorname{curl}(\mathbf{F})\cdot d\mathbf{S}$.

Proof: Shown in the notes on Blackboard.

*Note: The divergence theorem says the flux of **F** across the boundary of G is a triple integral over G. Notice the theme in Green's Theorem, The Divergence Theorem and Stokes' Theorem: some integral over a region is equal to another integral taken over the boundary of the region. This is similar to the way in which the Fundamental Theorem of Calculus was able to help find the area of region beneath a function by looking at the value of a different function at the endpoints.*

We now show some applications of Stokes' Theorem.

Example 5: Suppose that C is the triangle which is traced by moving from (0,0,1) to (1,0,0) to (0,1,0) to (0,0,1). Use Stokes' Theorem to find the exact value of the work done by the force field $\mathbf{F}(x,y,z)=(x+y^2)\mathbf{i}+(y+z^2)\mathbf{j}+(z+x^2)\mathbf{k}$ in moving a particle around the curve C.



We see that C encloses/bounds a surface σ.

σ is the surface given by

$x+y+z=1$

We select $\vec{n}=\langle 1,1,1\rangle$ so that

C is positively oriented.

OR $z=1-x-y$ so let $G=z-(1-x-y)$

& then $\vec{\nabla}G=\langle 1,1,1\rangle=\vec{n}$

Note that C is a pos. oriented, piecewise smooth, simple, closed curve that bounds σ. $\vec{F}$ has continuous partials.

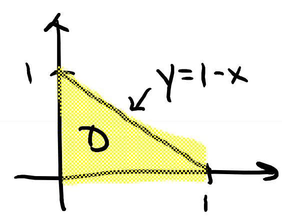
So by Stokes' Thm

$$\int_C \vec{F}\cdot d\vec{r} = \iint_\sigma \left(\text{curl}(\vec{F})\cdot\vec{n}\right)dS = \iint_D \text{curl}(\vec{F})\cdot\vec{\nabla}G\, dA$$

To compute this we need curl$(\vec{F})$ & the region D.

$$\text{curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y^2 & y+z^2 & z+x^2 \end{vmatrix} = \langle -2z, -2x, -2y\rangle$$

& $D=\{(x,y)\mid 0\le y\le 1-x \ \&\ 0\le x\le 1\}$



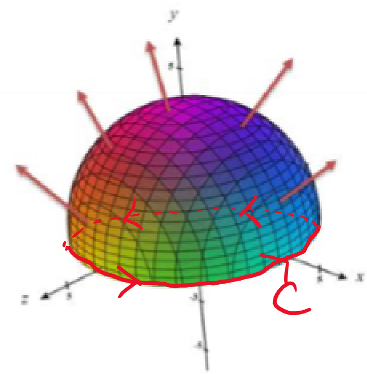
$$\iint_D \text{curl}(\vec{F})\cdot\vec{\nabla}G\,dA = \int_0^1\int_0^{1-x}\langle -2z,-2x,-2y\rangle\cdot\langle 1,1,1\rangle\,dy\,dx = \int_0^1\int_0^{1-x}\left(-2\left(\underline{1-x-y}\right)-2x-2y\right)dy\,dx$$

$$=\iint_D -2\,dA = -2\left(\tfrac{1}{2}(1)(1)\right) = -1$$

Example 6: Suppose that $\mathbf{F}(x,y,z)=\langle ze^{y}, x\cos(y), xz\sin(y)\rangle$. Suppose that σ is the hemisphere given by $x^2+y^2+z^2=16$ where $y\geq 0$, oriented in the direction of the positive y-axis. Find the exact value of $\iint_{\sigma} \operatorname{curl}(\mathbf{F})\cdot d\mathbf{S}$.

We note the boundary of σ is C which is a positively oriented, piecewise-smooth, simple, closed curve.

$\vec{F}$ has continuous partials.

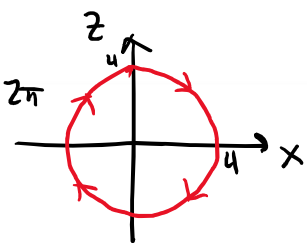


So by Stokes' Thm

$$\iint_{\sigma} (\operatorname{curl}(\vec{F})\cdot\vec{n})\,dS = \int_C \vec{F}\cdot d\vec{r}$$

We need to parameterize C

$$\vec{r}(t) = \langle \underbrace{4\sin(t)}_{x}, \underbrace{0}_{y}, \underbrace{4\cos(t)}_{z} \rangle \quad 0\leq t\leq 2\pi$$

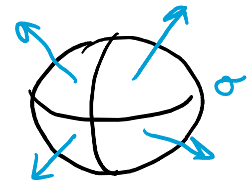


So $$\int_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} \vec{F}(\vec{r}(t))\,\vec{r}\,'(t)\,dt$$

$$= \int_0^{2\pi} \langle 4\cos(t), 4\sin(t), 0\rangle \cdot \langle 4\cos(t), 0, -4\sin(t)\rangle\,dt$$

$$= \int_0^{2\pi} 16\cos^2(t)\,dt = \ldots = 16\pi$$

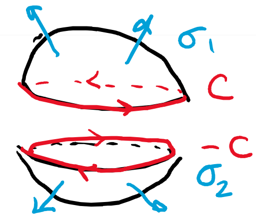
Example 7: Suppose σ is a sphere centered at the origin with positive orientation and $\mathbf{F}$ satisfies the hypotheses of Stokes' Theorem.



a) Is there a boundary curve of σ?

No!

b) Find $\iint_{\sigma} \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$



Note $\iint_{\sigma} \text{curl}(\vec{F}) d\vec{S} = \iint_{\sigma_1} \text{curl}(\vec{F}) d\vec{S} + \iint_{\sigma_2} \text{curl}(\vec{F}) d\vec{S}$

$= \int_C \vec{F} \cdot d\vec{r} + \int_{-C} \vec{F} \cdot d\vec{r}$

$= \int_C \vec{F} d\vec{r} + -\int_C \vec{F} d\vec{r}$

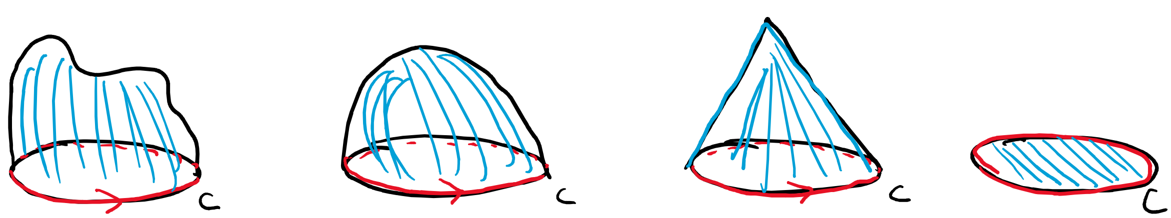
$= 0$

Example 8: Suppose $\mathbf{F}$ is a vector field whose components have continuous partials on all of three space. Suppose σ_1 is a surface that is bounded by the curve C such that σ_1 and C satisfy the hypotheses of Stokes' Theorem. Suppose that σ_2 is a surface that is bounded by the curve $-C$ such that σ_2 and $-C$ satisfy the hypotheses of Stokes' Theorem. If $\iint_{\sigma_1} \text{curl}(\mathbf{F}) \cdot d\mathbf{S} = \pi$, then what is the value of $\iint_{\sigma_2} \text{curl}(\mathbf{F}) \cdot d\mathbf{S}$?

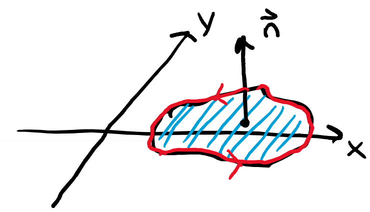
By Stokes' Thm

$\pi = \iint_{\sigma_1} \text{curl}(\vec{F}) d\vec{S} = \int_C \vec{F} \cdot d\vec{r} = -\int_{-C} \vec{F} d\vec{r} = -\iint_{\sigma_2} \text{curl}(\vec{F}) d\vec{S}$

So $\iint_{\sigma_2} \text{curl}(\vec{F}) d\vec{S} = -\pi$



Example 9: Suppose that σ is a surface in the xy-plane with upward orientation. Suppose that C is the boundary of σ. Suppose C and σ satisfy the hypotheses of Stokes' Theorem. Suppose that **F** is a vector field given by $\mathbf{F}(x,y,z)=\langle P(x,y),Q(x,y),0\rangle$ where **F** satisfies the hypotheses of Stokes' Theorem. Use Stokes' Theorem to find a formula for $\int_C \mathbf{F}\cdot d\mathbf{r}$.



By Stokes' Thm

$$\int_C \vec{F}\cdot d\vec{r} = \iint_\sigma \left(\text{curl}(\vec{F})\cdot\vec{n}\right)dS$$

σ is given by $z=0$ over a region D.

let $G=z-0$ & so $\vec{\nabla}G=\langle 0,0,1\rangle$

Also $\text{curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & 0 \end{vmatrix} = \left\langle 0,0,\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right\rangle$

So $\displaystyle\iint_\sigma \left(\text{curl}(\vec{F})\cdot\vec{n}\right)dS = \iint_D \left\langle 0,0,\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right\rangle\cdot\langle 0,0,1\rangle\, dA$

$$= \iint_D \left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dA$$

Note: We have obtained Green's Theorem from Stokes' Theorem in this example which is in line with our intuition that Stokes' Theorem is a generalization of Green's Theorem!